

Michele Biasutti

RICERCARE

for recorder (tenor & bass, one player) and signal processing

SCORE
2000

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Ricercare, is a late Renaissance and mostly early Baroque instrumental composition. Which have the preludial function to search out the key or mode of a following piece. The term is also used to designate a study that explores a technical device in playing an instrument. In the current piece *Ricercare* explore the application of some perceptive phenomena studied in the field of psychology of music, such as masking, space-time perception and some perceptive illusions on rhythm. The aim is to apply in music composition the results of this research, experimenting new formal developments. The data of the psychological research are utilized for transforming the musical material. In this way we can obtain new semantic meanings.

The sound figures presented in *Ricercare* are changing constantly: they deform, lengthen, transforming themselves in unexpected ways. The experimentation concerns the space-time organization using poly-rhythms, rhythms that are every time the same physically, but phenomenologically different, and the masking of simple rhythm's figures that constitutes a more complex context. Are considered the abilities in discriminating chords in the high and low registers and the limits of tonal perception. The perceptive paradoxes are utilized for generating and dissolving structural tensions.

Formally the piece is based on an ascending semi tone interval, using the tone G as tonal a center. The second interval expands itself gradually in a major third, creating perceptive games between minor and major third. The material develops gradually by rhythmic crescendos that step by step became sound clouds, composed of a myriad of notes in a third interval. The piece is articulated in sections, each of which on a specific perceptive phenomenon: minor and major third, timbre masking, rhythmic deformation, and limits in the discrimination of minor and major third.

The piece lasts eleven minutes and was first performed by Gianpaolo Capuzzo at Auditorium of Maddalene in Padova during the *Computer Art Festival 2000* on the 2nd June 2000 and awarded a Mention at the International competition *Musica Nova 2002* of Prague.

per flauto dolce (tenore e basso, un esecutore) ed elaborazione elettronica del suono

Il brano applica compositivamente alcuni fenomeni percettivi studiati nella psicologia della musica, sperimentando varie soluzioni formali. Tali fenomeni sono utilizzati per trasformare il materiale musicale, che assume in questo modo nuove connotazioni semantiche. Le figure sonore presentate sono in continua evoluzione: esse si deformano, si allungano, si dilatano, trasformandosi in elementi inattesi e imprevisi. La sperimentazione riguarda l'organizzazione spazio temporale con l'uso della poliritmia e con il mascheramento di semplici figure ritmiche che entrano a far parte di altre più complesse, che così conglobate non sono riconoscibili come elementi singoli. I paradossi percettivi sono utilizzati per creare e dissolvere le tensioni strutturali.

Formalmente il brano si basa su un intervallo di semitono ascendente, utilizzando come centro tonale la nota FA. L'intervallo di seconda si allarga fino ad arrivare a quello di terza maggiore, proponendo dei giochi percettivi tra terza maggiore e minore. Il materiale viene sviluppato gradualmente, con crescendo ritmici che sfociano in nuvole brulicanti di note, nelle quali i singoli toni perdono importanza in funzione della struttura complessiva. In questo contesto è perseguita l'asimmetria e l'indipendenza tra le voci. Il brano è articolato in sezioni, ognuna contraddistinta da un particolare approfondimento: terza maggiore e minore, mascheramento timbrico, deformazione ritmica.

Il brano, della durata di undici minuti, è stato eseguito da Gianpaolo Capuzzo in prima assoluta all'Auditorio delle Maddalene a Padova nell'ambito del *Computer Art Festival 2000* il 2 giugno 2000 e ha ottenuto la Menzione al concorso Internazionale *Musica Nova 2002* di Praga.

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Amplification

The recorder must be amplified with 1 microphone. You must use these sounds for making the signal processing. The sound of the recorder and of the processed sounds must be at equal volume through the loudspeakers.

Technical set-up

For the performance of the electronic part it needs:

1 delay with at least 4.0 sec.;

1 two voices harmonizer (the pitch change program of a digital sound processor) ranged from -12 to +12 halftones.

We suggest to use the Yamaha SPX 1000 as delay for his quality and any other device as harmonizer.

The delay has only one program:

DELAY: with 3.8 sec. delay on channel left and 4.0 sec. delay on channel right with a feedback of about 50%. Others changes concerning the feedback levels are reported in the score.

The harmonizer utilizes 5 programs:

ARM 1: pitch L: -0.5 halftones, delay L: 200 msec., pitch R: +0.5 halftones, delayR: 150 msec;
ARM 2: pitch L: -1.5 halftones, delay L: 200 msec., pitch R: +1.5 halftones, delayR: 150 msec;
ARM 3: pitch L: -12 halftones, delay L: 200 msec., pitch R: -7 halftones, delayR: 150 msec;
ARM 4: pitch L: -12 halftones, delay L: 200 msec., pitch R: +9 halftones, delayR: 150 msec;
ARM 5: pitch L: -12 halftones, delay L: 400 msec., pitch R: +12 halftones, delayR: 250 msec;

Note to the performance of the electronic part

In the score are reported only the signs that indicate the programs. For example “DELAY IN” at the beginning (staff 1), means that the delay must enter from that point. “ARM”, at staff 31, indicates that the harmonizer must enter with a fade-in from that point.

The harmonizer must be (in the mixer) only in the delay channel and not on the direct sound of the recorder.

Note that there is a measure that must not enter on the delay: you have to close only the input of the delay but not the output. This happened only one time in the score at staff 181 and it is written as “DELAY IN 0” when you have to close the input, and “DELAY IN 1”, when you have to open the input.

Ricercare

for recorder (bass and tenor, 1 player) and live electronics

per Gian Paolo Capuzzo

Michele Biasutti

Vivo ♩ = 100

Tenor Recorder

Live Electronics

ppp

DELAY IN 1, (4") feedback. 40%

The musical score is written for Tenor Recorder and Live Electronics in 4/4 time. It consists of 33 measures, grouped into systems of five measures each, with measure numbers 5, 10, 15, 18, 21, 24, 27, 30, and 33 marked at the beginning of their respective lines. The score includes various musical notations such as notes, rests, slurs, and triplets. Dynamics include *ppp*, *mf*, *p*, and *f*. Performance instructions for the live electronics include "DELAY IN 1, (4'') feedback. 40%" and "ARM 1 (in delay) +-0.5 IN".

39

pp

ARM 1+0.5 OUT

ARM 2 (in delay) +1.5 IN

ARM 1+-0.5 OUT

pp

ARM 2 (in delay) ± 1.5 IN

42

mf

mf

 f

51 
mf ARM 2 + - 1.5 OUT

 mf

ARM 2 + - 1.5 OUT

54

pp

3

3

 ${}^3_{pp}$

57

mf

rallentando

66

71 *a tempo*

ppp
(slap tongue) (after the delay)

74

p

77

80

mp

84

mf

87

89

f
ARM 1 (in delay) +0.5 IN

91

93

mf

95

97

ff

99

ARM 1 +0.5 OUT

101

ARM 2 (in delay) +1.5 IN

103

105

107

109

111

ARM +1.5 OUT

114

Delay feedback. 10%

117

4

123

ppp

128

p

131

mp

134

mf

137

f

140

f

143

p
ARM 3 (in delay) -12-7 IN

146

p

149

mp

152

mp

155 *mf*

158

161

164

166 *f*

168 *ff*

170

Delay feedback. 50%

172 *mf*

ARM 3 -12 -7 OUT

175 *mp*

178

take Bass Recorder

183

lasts about 30"

Più Lento ♩ = 50**Bass Recorder**

182

pp *very slow glissato*DELAY IN 0,
REV 10", ARM 3 (in reverb) -12 -7 IN

DELAY IN 1, REV 3.5", ARM 4 (in delay) -12 +9 IN

(harmonics of the fundamental F, rhythm free)

184

185

lasts about 8"

lasts about 8"

(multiphonics+ singing F)

188

lasts about 8"

lasts about 16"

take Tenor Recorder

(multiphonics+ singing A)

ARM 4 -12 +9 OUT

Tenor Recorder

190

pp

ARM 5 (in delay) -12 +12 IN

mf

(glissato lentissimo)

195

(glissato lentissimo)

f**pp**

ARM 5 -12 +12 OUT

(glissato lentissimo)

Presto

201

pp

202

203

mf

204

ppp

ARM 3 (in delay) -12 -7 IN

205

Staff 205: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4.

206

Staff 206: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *pp*

207

Staff 207: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *p*

208

Staff 208: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *mp*

209

Staff 209: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *mf*

210

Staff 210: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *f*

211

Staff 211: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4.

212

Staff 212: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *ppp*

213

Staff 213: Treble clef, key of D major. A continuous eighth-note melody starting on D4, moving up stepwise to D5, then down stepwise to D4. *fff*

Delay feedback. 10%

DELAY OUT O, REV 10"